

Physical Properties of the Period 3 Elements (Cambridge (CIE) A Level Chemistry): Revision Note

Describing Physical Properties of the Period 3 Elements

- Elements in the periodic table are arranged in order of increasing atomic number and placed in vertical columns (**groups**) and horizontal rows (**periods**)
- The elements across the periods show **repeating patterns** in chemical and physical properties
- This is called **periodicity**

Arrangement of elements in the Periodic Table

PERIODIC TABLE OF THE ELEMENTS

A diagram of a periodic table with a grey background. A box labeled 'KEY' has an arrow pointing to the periodic table. Helium (He) is highlighted in a yellow box. Labels with arrows point to its properties: 'MASS NUMBER' points to the top-left '4', 'ELEMENT SYMBOL' points to the top-right 'He', 'ATOMIC NUMBER' points to the bottom-left '2', and 'NAME' points to the bottom-right 'HELIUM'. To the right of the periodic table, five colored boxes represent states at room temperature: 'METALS' (blue), 'NON-METALS - SOLID' (purple), 'NON-METALS - GAS' (yellow), 'LIQUIDS' (red), and 'METALLOID' (green). The text 'AT ROOM TEMPERATURE' is centered below these boxes.

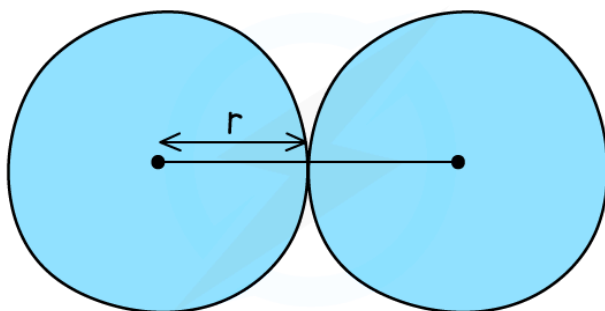
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Elements are arranged by increasing atomic number from left to right

Atomic radius

- The **atomic radius** is the distance between the nucleus and the outermost electron of an atom
- The atomic radius is measured by taking two atoms of the same element, measuring the distance between their nuclei and then halving this distance
- In **metals**, this is also called the **metallic radius** and in **non-metals**, the **covalent radius**

Atomic radius



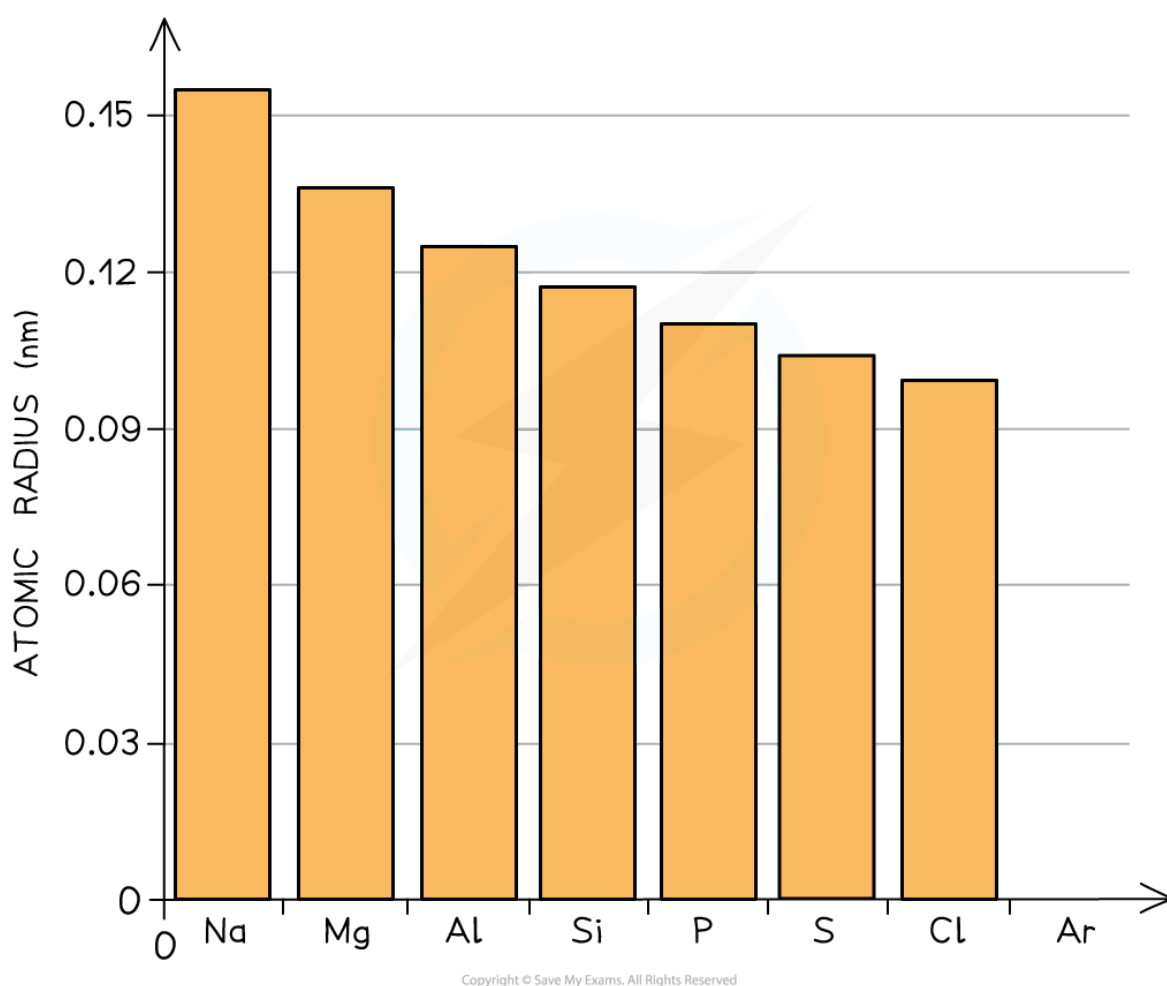
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The atomic radius gives a measure of the size of atoms

Atomic radii of Period 3 elements table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Atomic radius (nm)	0.157	0.136	0.125	0.117	0.110	0.104	0.099	-

Graph of atomic radii across Period 3



There is a decrease in atomic radii of Period 3 elements across the period

- Across the period, the atomic radii decrease
- This is because the number of protons (**the nuclear charge**) and the number of **electrons** increases by one every time you go an element to the right
- The elements in a period all have the same number of shells (so the **shielding effect** is the same)
- This means that as you go across the period the nucleus attracts the electrons more **strongly pulling them closer** to the nucleus
- Because of this, the atomic radius (and thus the size of the atoms) **decreases** across the period

Ionic radius

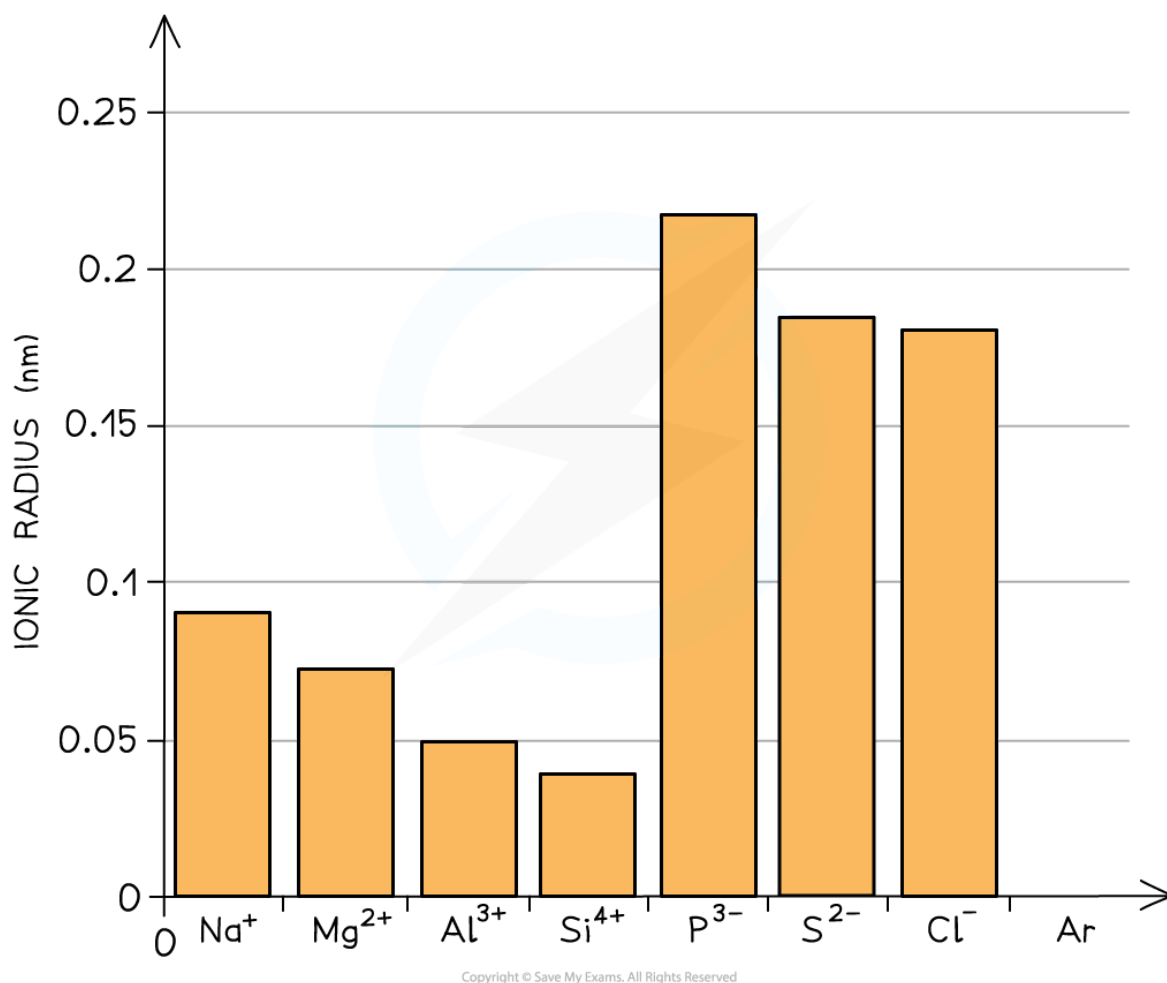
- The **ionic radius** is the distance between the nucleus and the outermost electron of an ion
- Metals produce positively charged ions (**cations**) whereas nonmetals produce negatively charged ions (**anions**)

- The **cations** have lost their valence electrons which causes them to be much **smaller** than their parent atoms
 - This is because there are less electrons, which also means that there is less **shielding** of the outer electrons
- Going across the period from Na^+ to Si^{4+} the ions get **smaller** due to the **increasing nuclear charge** attracting the outer electrons in the **second principal quantum shell** nucleus (which has an increasing atomic number)
- The **anions** are larger than their original parent atoms because each atom has gained one or more electrons in their **third principal quantum shell**
- This increases the repulsion between electrons, while the **nuclear charge** is still the same, causing the electron cloud to spread out
- Going across P^{3-} to Cl^- , the ionic radii **decrease** as the nuclear charge increases across the period and fewer electrons are gained by the atoms (P gains 3 electrons, S 2 electrons and Cl 1 electron)

Ionic radii of ions of Period 3 elements table

Period 3 ion	Na^+	Mg^{2+}	Al^{3+}	Si^{4+}	P^{3-}	S^{2-}	Cl^-	Ar
Ionic radius (nm)	0.095	0.065	0.050	0.041	0.212	0.184	0.181	No data

Graph of ionic radii across Period 3 ions



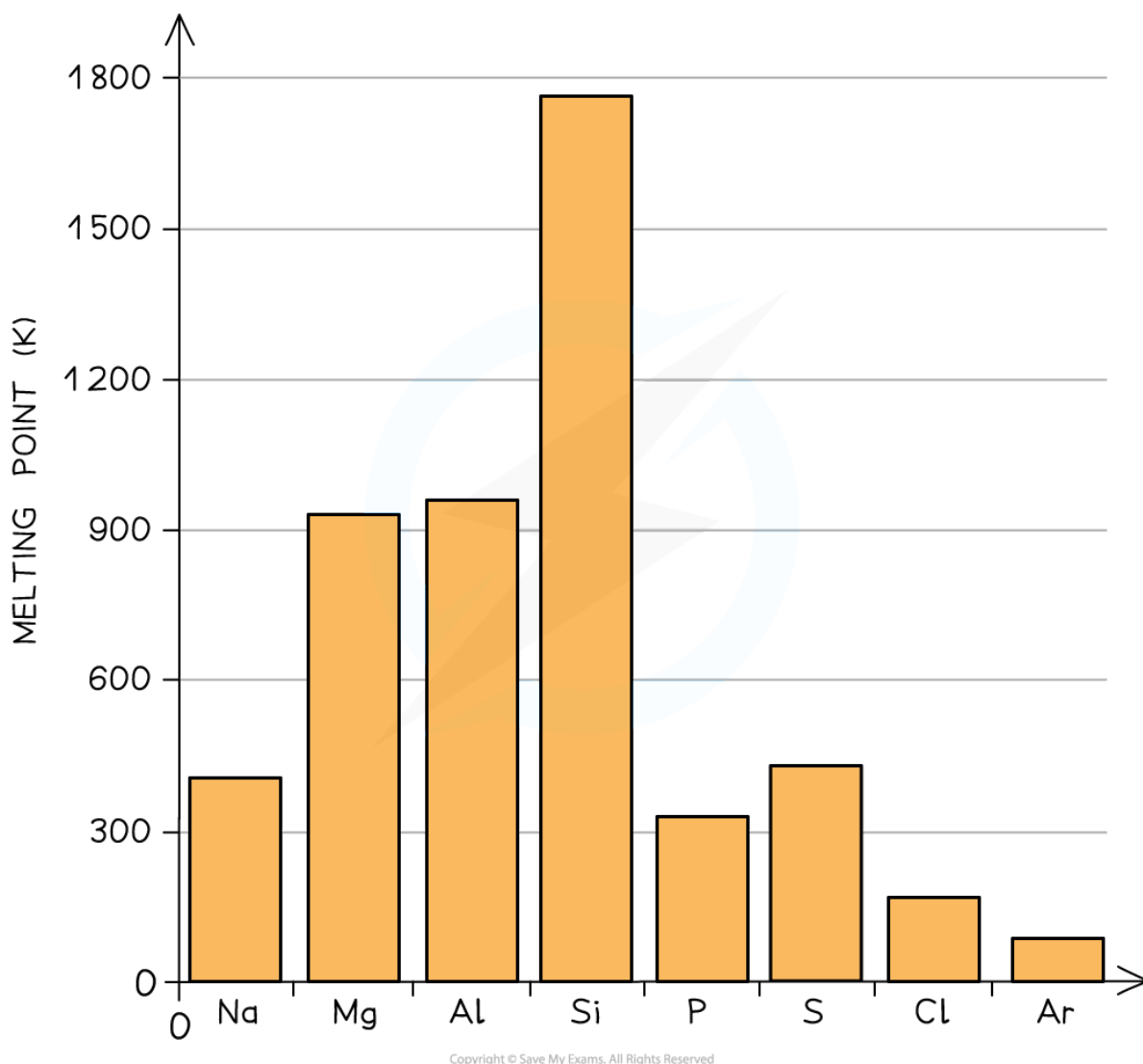
Ions of Period 3 elements with increasing positive charge (metals) and increasing outer electrons across the period

Melting point

Melting points of the elements across Period 3 table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Melting point (K)	371	923	932	1683	317	392	172	84

Graph of melting points across Period 3



There is a general increase in melting point from Na to Si, followed by a sharp drop to the lower melting points of P to Ar

- A general increase in **melting point** for the Period 3 elements up to silicon is observed
- **Silicon** has the highest **melting point**
- After the Si element, the melting points of the elements **decrease** significantly

Electrical conductivity

- **Electrical conductivity** refers to how well a substance can conduct **electricity**
- Unlike the melting points, the electrical conductivity of the Period 3 elements shows a clear trend
- Going across the period, the electrical conductivity of the elements **decreases** significantly
 - Initially there is an increase in the electrical conductivity from Na to Al and then this decreases across the remaining elements

Trends in electrical conductivity across Period 3 table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Electrical conductivity (S m ⁻¹)	0.218	0.224	0.382	2×10^{-10}	10^{-17}	10^{-23}	-	-